High-Power Inductively Coupled Impulse Sputtering Glow Plasma

Ken Yukimura, Hisato Ogiso, Shizuka Nakano, and Arutiun P. Ehiasarian

***Abstract*—A high-power pulsed glow plasma consisting of ionized metallic species was generated at the sputter target without the use of an externally applied magnetic field. An inductively coupled plasma was produced with a radio frequency (RF) of 200 kHz, and the plasma production was in a burst mode. A pair of titanium electrodes as the sputter target was placed near the RF argon plasma source. A pulsed direct-current (dc) voltage with negative polarity was applied to the target. The metallic plasma, which is referred to as the target plasma, had a density of 10^{13}–10^{14} cm^{-3}. The electrical characteristics of the target plasma were dependent on the timing of the application of the pulsed dc voltage. The target voltage and the current through the target change based on the plasma density, and the consumed power reaches a maximum when the impedance matches the target plasma circuit, i.e., when the plasma impedance is consistent with the circuit resistance in the target plasma circuit. Optical emission spectra of the target and RF plasma regions were recorded, and titanium and argon ions were observed. Titanium ions were generated within 10 μs after the application of the negative pulse voltage and then diffused into the RF plasma region. The production region for titanium ions occurred within approximately 10 mm of the target surface, and diffusion occurred during the application of the dc pulse in the RF plasma region. However, diffusion was substantially limited to within approximately 35 mm from the target surface. The optical emission intensity of the titanium ions is dependent on the power consumed in the target plasma, and hence, a maximum intensity can be obtained at the maximum power consumption, at which impedance matching is satisfied. The parameters that influence the emission intensity of the metallic species include the plasma density and the sputtering yield.**

***Index Terms*—Glow discharge, high-power impulse magnetron sputtering (HIPIMS), high-power pulsed magnetron sputtering (HPPMS), high-power pulsed sputtering (HPPS), radio-frequency (RF) plasma.**

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I. Introduction

HIGH-POWER pulsed glow plasma has a significant potential for uses in surface modifications of industrial components and biomaterials. The instantaneous electrical power is on the order of several tens to a few megawatts. Materials can be characterized on the basis of their electrical properties, i.e., conductivity or insulation ability in semiconductor devices, and their tribological properties, i.e., friction and wear. Metallic species subjected to high-power plasma are highly ionized, and the glow plasma density is on the order of 10^{13}–10^{14} cm^{-3} even at a pressure of 1 Pa. There have been several types of glow plasma systems developed. A magnetron sputtering system operated by a pulsed high-power plasma was first proposed near the end of the 1990s by Kouznetsov *et al.* [1]. This method is referred to as high-power impulse magnetron sputtering [2] or high-power pulsed magnetron sputtering [3]. A high-power-consumption glow plasma can be also generated by a Penning discharge [4]–[6] and by a pulsed inductively coupled plasma [7]. These methods typically involve a power value at the target surface on the order of 100–1000 kW for a discharge current density of 1–100 A/cm^2 and a discharge voltage in the range of 400–1000 V.

Plasma generation and maintenance of the pulsed magnetron sputtering and the Penning discharge that were aforementioned are based on the use of an externally applied magnetic field to confine charged particles, particularly electrons, to the plasma. The electric and magnetic fields facilitate ionization of the sputtered species in the plasma. Conversely, when a magnetic field is used, charged particles may not be linearly accelerated along the electric field. This effect causes not only random ionic motion but also nonconformal ion implantation and deposition during material processing. Thus, the generation of a metallic plasma even in the absence of a strong magnetic field is desirable for enhanced uniformity of film preparation and conformal ion implantation.

A high-power inductively coupled plasma (ICP) has been previously proposed [7]. A solenoid-type inductor is used to generate a gas plasma, and the current through the solenoid can reach 100 A, which produces high-density argon ions. The plasma density is on the order of 10^{13}–10^{14} cm^{-3}. The radio-frequency (RF) power is generated by the resonance between a charged capacitor and an inductor, and an impedance matching circuit is not used in RF plasma generation. Thus, the plasma generation circuit is simplified. The RF frequency is approximately 200 kHz, and a burst voltage applied to the inductor is 3 kV or higher.

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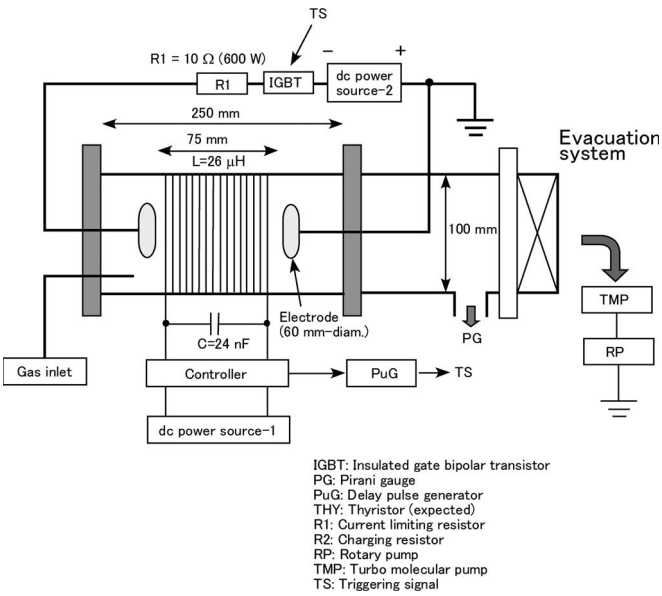

Fig. 1. Experimental apparatus.

By applying a pulsed direct-current (dc) voltage with negative polarity to the sputter target placed in the RF plasma, energetic ion bombardment efficiently sputters the target material, which is ionized in the plasma. A plasma containing charged metallic species is generated around the target, which is referred to as the target plasma, and the plasma density reaches 10^{13}–10^{14} cm^{-3}. The current through the target is several tens of amperes for target voltages of several hundreds of volts. The instantaneous power consumed in the target plasma is on the order of several tens of kilowatts, which corresponds to a power density over 1 kW/cm^2 at the target.

A high-power glow discharge plasma was previously generated in the 1970s by combining a pulsed dc power source with an RF power source (as described by Malkin [8]). In this case, the production of metal ions was not the goal of the study.

This paper is a continuation of a previous research study [7]. In this paper, fundamental electrical and optical emission characteristics, as well as the production of metallic species, are discussed on the basis of circuit impedance matching, i.e., the target plasma maintains a high density when generated within the RF plasma because abundant energetic argon ions bombard the target. The parameters that influence titanium ion production are also discussed. The plasma density, which is based on the target current, and the sputtering yield, which is based on the target voltage, are the main factors influencing the generation of metallic ions. In addition, the consumed power is closely associated to sputtering and ionization of the metallic species. The highest emission intensity originating from the metallic ions is obtained under conditions of impedance matching when the highest power is consumed at the target.

II. Experimental Procedure

A. Whole Structure

Fig. 1 shows the experimental apparatus comprised of a solenoid coil with an RF antenna mounted on the outside wall of a containment vacuum chamber with a diameter of 100 mm and a length of 250 mm. A pair of disk-shaped titanium electrodes was placed near the RF plasma region. The diameter and thickness of each electrode were 60 mm and 10 mm, respectively. The electrodes were used as sputter targets that eject metallic species when subjected to energetic argon ion bombardment. No externally applied magnetic field was used to sputter the electrodes, thus simplifying the apparatus.

B. RF Plasma Source

The plasma source consisted of a 24-nF capacitor and an 18-turn solenoid inductor with an inductance of 26 μH, which were used to generate an RF ICP. The length of the inductor coil was approximately 50 mm. The capacitance and inductance were resonant with a frequency of approximately 200 kHz. Thus, no impedance matching circuit was used. The values of the capacitance and inductance can be changed, and the resonance frequency can be adjusted within the range of 100–300 kHz. A 200- to 260-μs-long burst of RF power was used to produce an argon plasma. The number of RF pulses and the repetition rate were regulated by a controller set between the resonance component and the dc power source-1 shown in Fig. 1; there were typically 50 cycles for one burst pulse at a rate of 60 Hz.

C. Sputter Target

The sputter target had a negatively pulsed bias to attract ions from the RF plasma to then generate the target plasma. Thus, the pulsed negative voltage accelerates the argon ions generated in the RF ICP toward the sputter target cathode. The gap length was 75 mm. One of the electrodes was configured as the cathodic sputtering target, and the second electrode was grounded to complete a target plasma circuit.

D. Sputtering Sequence

Sputtering of the metallic species and its ionization were initiated by the generation of a 200-kHz RF argon ICP, followed by the application of a pulsed negative voltage to a pair of electrodes at a designated time after the RF burst was produced. This time is referred to as the delay time, which is defined as the time between the beginning of the RF voltage output and the turn-on time of the insulated-gate bipolar transistor (IGBT). The target material is sputtered by the bombardment of the target electrode by energetic argon ions. A discharge containing the ionized metallic species is generated at the target. The discharge is referred to as the target plasma. The sputtered metallic atoms are highly ionized in the target plasma, and the ions diffuse in the RF plasma.

The RF voltage signal triggers a pulse generator (PuG) to apply a signal to the gate circuit to turn on the IGBT. The IGBT acts as a switch that closes the target circuit and is shut off when the appropriate signal is received. Thus, the ON-time duration of the IGBT is determined by the PuG signal. Because the PuG also controls the start time of the voltage application, the generation of the RF ICP and the target plasma is controlled by a PuG signal.

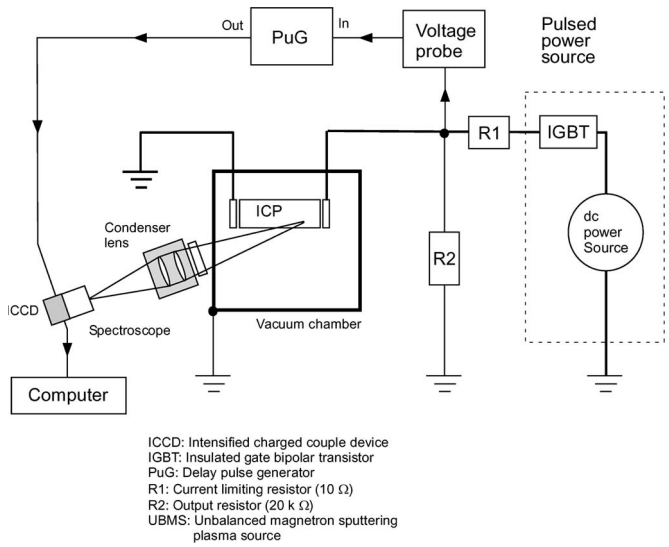

Fig. 2. Observation of optical emission spectrum using a spectroscope and an ICCD camera (the observatory time is controlled by the output voltage of the PuG).

To attract ions into the RF plasma, the electrodes are connected to a pulsed dc power source, i.e., source-2, via a current-limiting 10-Ω resistor, i.e., R1, (rated power: 600 W) and the IGBT switch. The pulsed-power source-2 has a 300-A current capability with a maximum negative polarity output voltage of 1.5 kV. The target voltage is applied at a designated time since the RF voltage application, which is referred to as a delay time.

In this paper, the source voltage is the output voltage of dc power source-2 shown in Fig. 1.

E. Gas Supply and Evacuation

The ambient gas was 99.99% pure argon gas; using a mass flow controller, the gas was supplied by a gas inlet positioned on the opposite side of the evacuation units. A typical flow rate was 20 standard cubic centimeters per minute (sccm) , and the gas pressure inside the glass chamber was varied from 0.1 to 20 Pa by varying the flow rate.

The pumping unit consisted of turbomolecular and rotary pumps with volume flow rates of 350 and 100 L/s, respectively. The pressure was measured at the side of the pumping unit using Pirani and ion gauges.

F. Observation of Optical Emission Spectra

Fig. 2 shows a schematic for the observation of optical emission spectra of the glow plasma. Emission of visible light and ultraviolet light was observed using a spectroscope with a gated intensified charge-coupled device (ICCD, Andor, model DH510) to record the emission intensities. The focus points for the observation of optical emission spectra were adjusted from the surface to the inside portion of the RF plasma using a collimator lens placed in front of the spectroscope. The emission intensity of the spectrum signal was recorded by a computer. The focal length of the spectrometer was 75 mm with a diffraction grating of 300 grooves/mm (Blaze wavelength = 300 nm). The gate time of the ICCD was 100 ns, and the shutter time (exposure time of the light emission) was 0.1 s. The optical emission spectra for ten shots were observed at a repetition rate of 60 Hz to improve the signal-to-noise ratio. The slit width at the plasma inlet was 0.05 mm. The ICCD gate was opened by a PuG signal triggered by a so-called transistor–transistor logic (TTL) level voltage signal at a designated time for the target voltage. Thus, the time of an observation was determined by the output signal of the PuG.

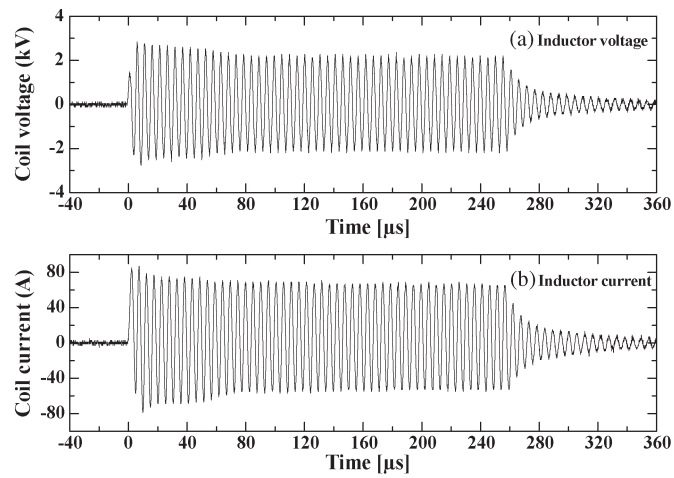

Fig. 3. Waveforms of the 225-kHz RF voltage across the (a) inductor and the (b) current through the inductor for the width of burst pulse of 260 μs (initial RF source voltage: 2.8 kV, argon gas pressure: 4.1 Pa).

III. Experimental Results and Discussion

A. RF ICP Initiation and Ion Extraction

Waveforms of the voltage across the inductor and the corresponding current through the inductor are shown in Fig. 3(a) and (b). The chamber was filled with argon gas at a pressure of 4 Pa. The burst-mode applied voltage lasted approximately 260 μs to generate a train of approximately 60 sinusoidal cycles at a frequency of approximately 225 kHz. The current phase is 90° from the applied voltage. Approximately seven cycles (approximately 30 μs) are required to initiate an RF plasma. The voltage and current prior to plasma initiation are 2.7 kV and 80 A, respectively. After RF plasma initiation, the target current decreases to approximately 68 A from the initial current of 80 A, and the inductor voltage decreases from 2.7 kV to approximately 2.3 kV. Current damping occurs because the plasma is coupled to the solenoid-type inductor, and the plasma acts as a one-turn coil. Thus, the RF current and the plasma become a transformer coupling. The starting time of the current damping is defined as the RF plasma ignition time.

The ignition time of the RF plasma as a function of the argon gas pressure, where the initial RF peak voltage is 3.2 kV, is shown in Fig. 4. The minimum ignition time occurs at approximately 26 μs and 2 Pa. A gradual increase in the ignition time is observed in the pressure range from 0.1 to 2 Pa, and the ignition time significantly increases when the pressure increases above 2 Pa. Overall, the ignition time exhibits a characteristic U-shape in relation to the pressure. The U-shape characteristic shows that a degree of the electron production is dependent on the surrounding gas pressure.

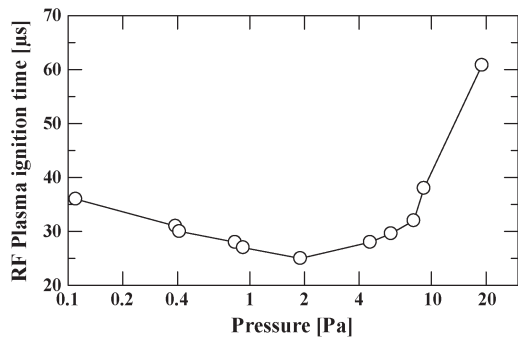

Fig. 4. RF plasma ignition time as a function of the argon ambient pressure (initial RF source voltage: 2.8 kV).

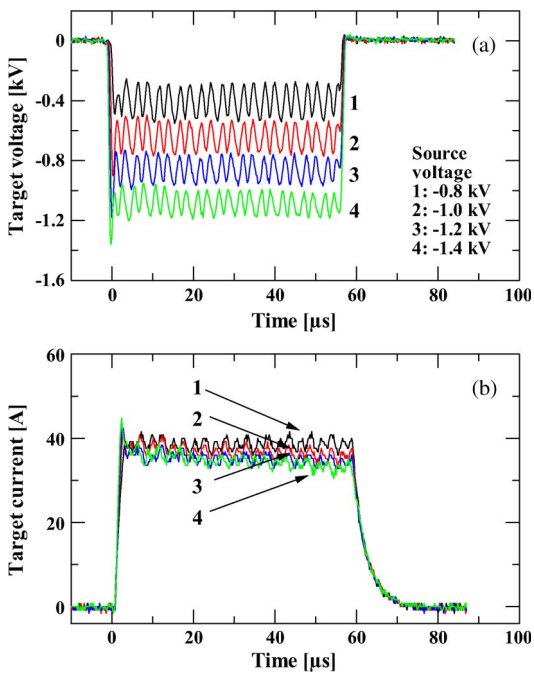

Fig. 5. Waveforms of the (a) target voltage and the (b) current through the circuit for source voltages of −800, −1000, −1200, and −1400 V (a 200-kHz RF source voltage: 3.2 kV (peak value); a width of the RF burst pulse: 200 μs; a delay time: 100 μs; an inserted series resistance: 10 Ω; a pulse repetition rate: 60 Hz; and an argon gas pressure: 1 Pa).

When a pulsed dc voltage is applied to the target, ions are drawn from the RF plasma to sputter the target material. The voltage application time is thus classified into three regions. Case-1 refers to the time during RF plasma generation (see Fig. 5), case-2 refers to the time at the end of the burst pulse (see Fig. 6), and case-3 refers to the time in the after-glow region (see Fig. 7). In each time region, the voltage and current characteristics change. The electrical characteristics were investigated under an RF burst pulse duration value of 200 μs.

For case-1, the waveforms of the target voltage and the corresponding current through the target are shown in Fig. 5(a) and (b), respectively. The source voltage was varied from −0.8 to −1.4 kV, the delay time was 100 μs, and the argon gas pressure was fixed at 1 Pa. Although an RF signal-based component was superimposed as noise on both waveforms of the voltage and the current, the dc component of the voltage at each applied voltage does not change during the pulse. The

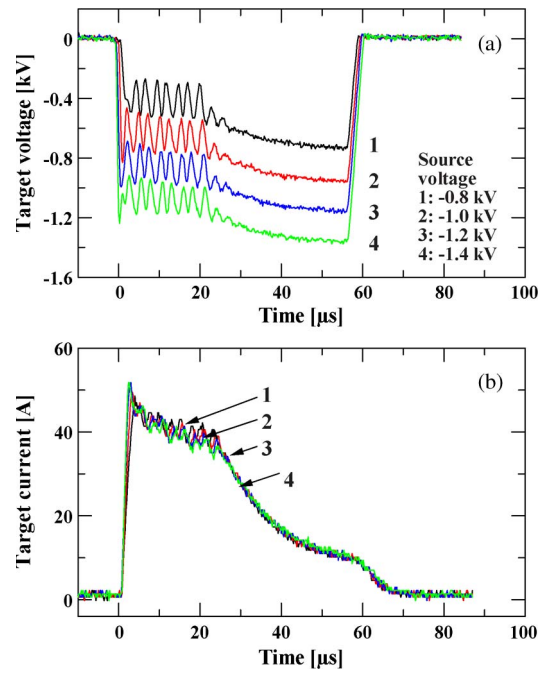

Fig. 6. Waveforms of the (a) target voltage and the (b) current through the circuit for source voltages of −800, −1000, −1200, and −1400 V (a 200-kHz RF source voltage: 3.2 kV (peak value); a width of the RF burst pulse: 200 μs; a delay time: 185 μs; an inserted series resistance: 10 Ω; a pulse repetition rate: 60 Hz; and an argon gas pressure: 1 Pa).

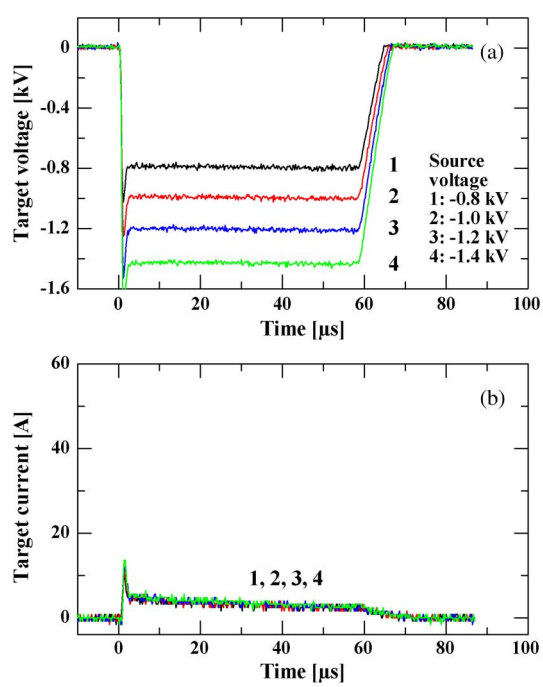

Fig. 7. Waveforms of the (a) target voltage and the (b) current through the circuit for source voltages of −800, −1000, −1200, and −1400 V (a 200-kHz RF source voltage: 3.2 kV; a width of the burst pulse: 200 μs; a delay time: 295 μs; an inserted series resistance: 10 Ω; a pulse repetition rate: 60 Hz; and an argon gas pressure: 1 Pa).

average target voltage was observed to be lower than the source voltage due to a voltage drop of approximately 400 V at the 10-Ω current-limiting resistor denoted as R1 (see Fig. 1). The plasma impedance, which is equal to the target voltage/current through the target, changes from approximately 10 to 25 Ω at

the applied target voltages of -0.8 to -1.4 kV. The current through the target was identical, although the source voltage was varied, i.e., the current is the so-called ion saturation current.

The plasma density is estimated to be approximately 4×10^{13} cm^{-3} at an assumed electron temperature value of 5 eV [9]. In the actual experiment, the generated ion plasma may include both sputtered titanium species and argon ions, which can be observed in the optical emission spectra. However, the density ratio of these species was not determined in this study.

The instantaneous power consumed in the target plasma is proportional to the applied target voltage because the target current is identical. As shown in Fig. 5, the consumed power ranges from 15 to 42 kW for initial source voltages of -0.8 to -1.4 kV, corresponding to power densities of approximately 0.54–1.5 kW/cm^2 at the target, respectively. These values are 5- to 15-fold higher than those produced using the conventional ionized sputtering method [10], [11].

For case-2, waveforms of the target voltage and the current through the target at a delay time of approximately 185 μs are shown in Fig. 6(a) and (b). The RF burst ends at 20 μs, after which the plasma is in an extinguishing stage, i.e., in a deionization stage, in which a significant amount of the charged particles is lost; in terms of the electrical characteristics, the current decreases and the voltage increases to return to the original insulating state. However, the current remains the same and is identical even at varying initial source voltages. Significant deionization occurs in this stage, and the current is reduced to almost 25% of the original value at 60 μs, which is 40 μs after the end of the burst pulse. Thus, the charged particles are swiftly lost by a recombination after the termination of the RF power.

For case-3, the waveforms of the target voltage and the target current at a delay time of approximately 295 μs for a variety of initial target voltages are shown in Fig. 7. The plasma is in what is referred to as the after-glow region, and the extracted ion current is less than 5 A, although the current is identical even at varying target voltages. This current is also an ion saturation current. The observed target voltage is the same as the source voltage because the plasma impedance ranges from 160 to 280 Ω, and this value is 16–28 times higher than R1 at 10 Ω. The plasma density in the after-glow region is estimated to be approximately 4.4×10^{12} cm^{-3}; thus, the target plasma density is one order of magnitude lower than that of the plasma.

In summary, argon ions are accelerated toward the target and sputter the target material. The sputtered species are ionized in the target plasma, and the target voltage is determined by the impedance of the target plasma.

Waveforms of the target voltage and the target current for a variety of RF voltages between 2.3 and 4.2 kV at a fixed delay time of 105 μs, for which the duration of the burst pulse is approximately 205 μs, are shown in Fig. 8(a) and (b), respectively. Similar to the conditions in Fig. 5, the source voltage is applied during RF plasma generation. The observed target current increases as the RF voltage increases. Thus, the plasma density is proportional to the RF voltage. Fig. 9 shows the plasma density as a function of the target current, in which the observed current is an ion saturation current at an assumed electron temperature value of 5 eV [9]. The plasma is also assumed to be uniform inside the glass chamber. The conventional magnetron sputtering discharge is enhanced by an ICP plasma, and the power density in the plasma region is approximately 25.4 W/cm^2 [10]; in this study, the plasma density is approximately three orders of magnitude higher because the power density in the plasma region is approximately 400 W/cm^2 for 32 kW of instantaneous power, which is approximately 20 times higher than the enhanced ICP plasma that was previously reported. Thus, the plasma density significantly increases with the amount of consumed power.

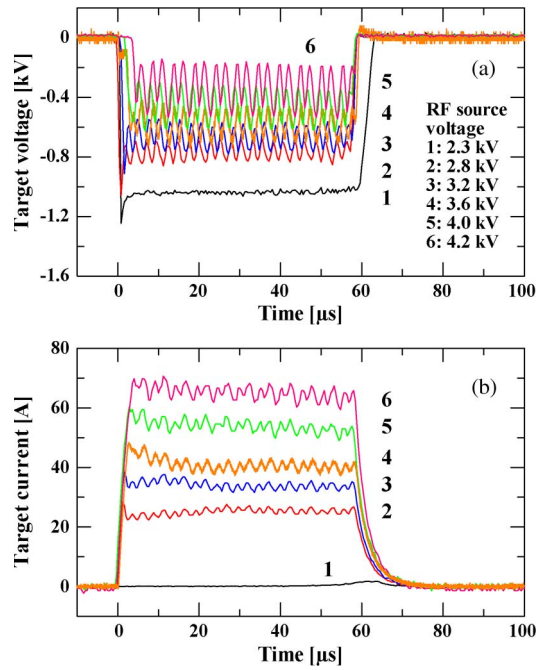

Fig. 8. Waveforms of the (a) target voltage and the (b) current through the circuit for a 200-kHz RF source voltages of 2.3, 2.8, 3.2, 3.6, 4, and 4.2 kV (peak values) (a target voltage: -1.2 kV; a width of the RF burst-pulse: 200 μs; a delay time: 100 μs, an inserted series resistance: 10 Ω; a pulse repetition rate: 60 Hz; and an argon gas pressure: 1 Pa).

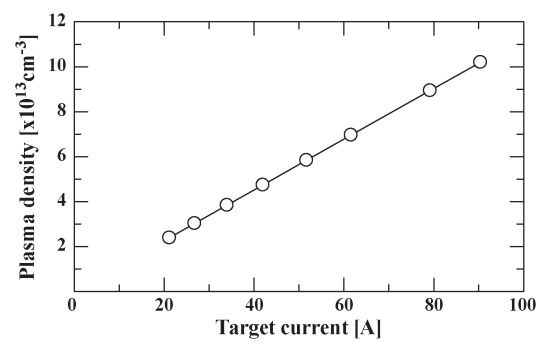

Fig. 9. Plasma density as a function of the target current, where a peak RF source voltage is varied from 2.3 to 4.2 kV (peak values) (the experimental conditions are identical to Fig. 8).

The electrical characteristics as a function of the RF output voltage are shown in Fig. 10(a)–(c). The target current increases, and the target voltage decreases, as the RF voltage increases. This result indicates that the target plasma impedance decreases as the RF voltage increases due to the increase in the concentration of argon ions that act as seed particles to sputter

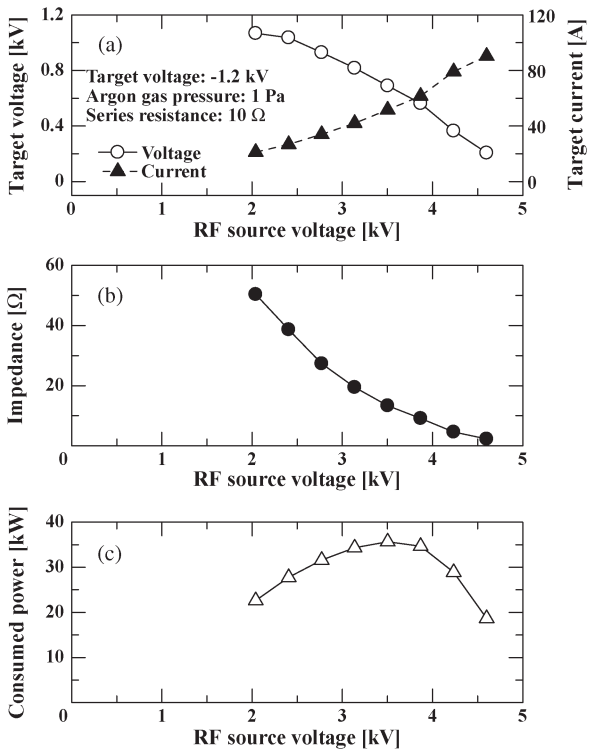

Fig. 10. (a) Target voltage and current through the target, (b) impedance of target plasma, and (c) power consumed in the target plasma as a function of the RF source voltage (the experimental conditions are identical to Fig. 8).

the target material. As shown in Fig. 10(b), the impedance ranges from 50 to 1 Ω for RF voltages from 2 to 4.6 kV, respectively. Fig. 10(c) shows the power consumed in the target plasma as a function of the RF output voltage. The power that is experimentally obtained satisfies the circuit conditions for impedance matching.

The power consumed in the target plasma has a peak maximum at an RF voltage of 3.6 kV, which corresponds to an impedance value of 10 Ω. Thus, the power has a peak maximum under conditions of impedance matching in the target plasma circuit [12]. The circuit consists of a power source with internal impedance Z_i and output voltage V_0. A series resistor, i.e., Z_s, is inserted between the voltage source output and the plasma as a load on the voltage source, where the plasma impedance is Z_T. Z_c can be treated as a sum of the internal impedance or as the output impedance of the power source ($Z_c = Z_i + Z_s$).

The power, i.e., P, consumed in the glow plasma can be represented as

$$P = i \cdot V_0 - i^2 \cdot Z_c \tag{1}$$

where the power is restricted to only positive values [12]. Maximum power P_m can be obtained at $Z_c = Z_T$. This condition satisfies the impedance matching condition, in which the current flow is $i = V_0/(2Z_c)$. Thus, maximum power can be obtained at the impedance matching condition, $Z_T = Z_c$. The consumed power can be normalized as

$$\frac{P}{V_0^2/(4Z_c)} = \frac{i}{V_0/(2Z_c)} \cdot \left\{2 - \frac{i}{V_0/(2Z_c)}\right\}. \tag{2}$$

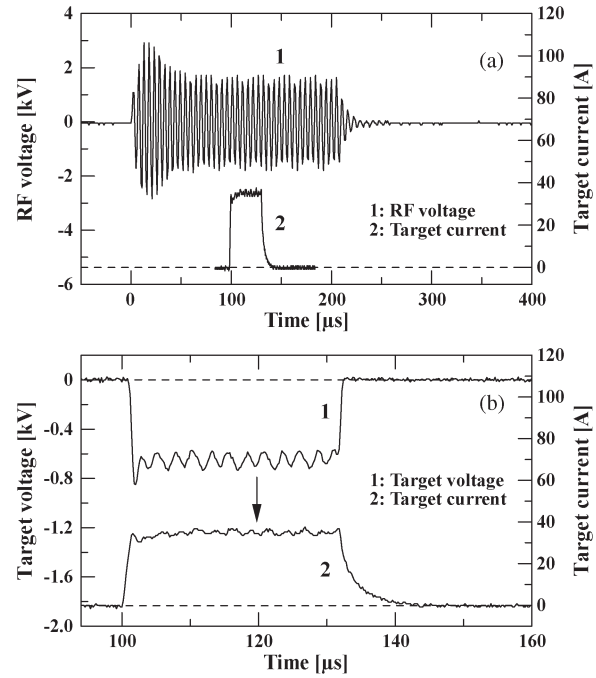

Fig. 11. Waveforms of the RF burst pulse voltage across the inductor and the (a) target current, and the target voltage and the current through the target (Delay time: 100 μs; RF voltage: 2.8 kV (peak value); argon gas pressure: 0.9 Pa; and source voltage applied to the target: -1 kV).

B. Relationship Between Optical Emission Intensity and Impedance Matching of the Circuit

Optical emission spectra from the target plasma was observed at a delay time of 100 μs after the RF voltage application. Fig. 11(a) and (b) shows waveforms of the RF voltage and the target current through the target plasma and the target voltage and current, respectively. The duration of the 200-kHz RF burst was approximately 205 μs, and the duration of the target voltage was approximately 30 μs. The argon pressure was 1 Pa. The target voltage and the current were observed to be approximately 600 V and 35 A, respectively. Hence, the instantaneous power is approximately 21 kW. Because the RF plasma is initiated at approximately 27 μs, the target is immersed in the RF plasma when the pulse voltage is applied to the target [see Fig. 11(b)].

Fig. 12 shows an example of optical emission spectra observed at the target surface at 20 μs after the pulsed voltage was applied to the target [this time, it is shown as an arrow in Fig. 11(b)]. Titanium- and argon-based spectra are observed. In particular, the optical emission spectra originating from a singly ionized titanium with wavelengths of 323.45 nm (No. 1, shown in Fig. 12) and 334.94 nm (No. 2) are significant, and the titanium atomic spectrum (emission from an excited state of titanium) is observed at 375.29 nm (No. 6). The argon-based spectra are observed at 434.81 nm (No. 11) and 442.60 nm (No. 12), as well as at wavelengths ranging from approximately 430 to 500 nm. These wavelengths were specified using optical emission spectrum data [13]. These results confirm that titanium species are sputtered and ionized in the plasma and that the argon ions are efficiently accelerated by the negative bias voltage applied to the sputter target.

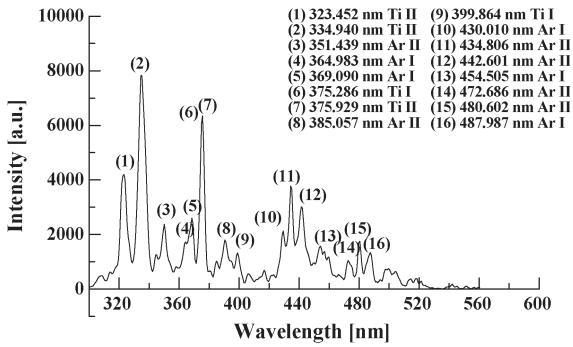

Fig. 12. Optical emission spectrum of target plasma at the target surface at 20 μs since the voltage application of the sputter target (the experimental conditions are identical to Fig. 11).

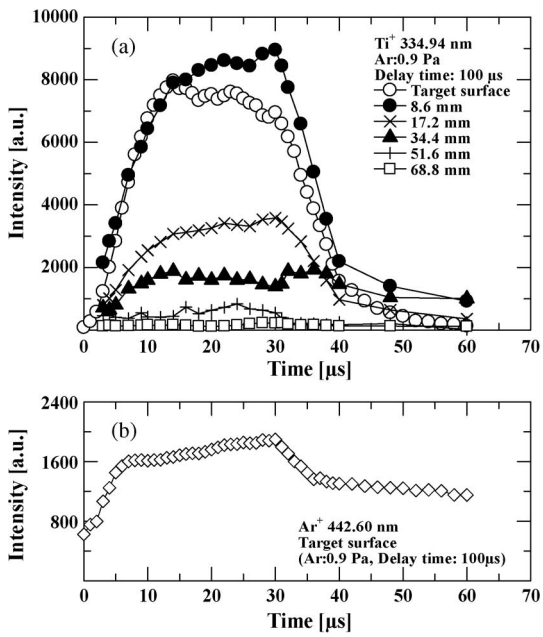

Fig. 13. Intensities of optical emission spectra of (a) singly ionized titanium ions (334.94 nm) and (b) singly ionized argon ions (442.60 nm) as a function of the time after the pulsed voltage application (the experimental conditions are identical to Fig. 11).

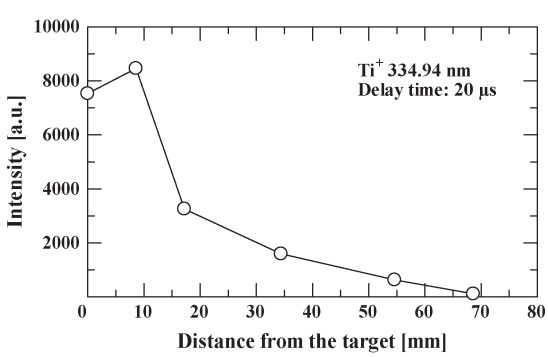

Fig. 14. Intensities of optical emission spectra of singly ionized titanium ions (334.94 nm) as a function of the distance from the sputter target at 20 μs since the pulsed voltage application (the experimental conditions are identical to Fig. 11).

Fig. 13(a) and (b) shows the temporal behavior of optical emission spectra for singly ionized titanium and argon ions at wavelengths of 334.94 and 442.60 nm, respectively. Fig. 14 shows the titanium ion spectrum (334.94 nm) intensity as a function of the distance from the target 20 μs following the start of the pulse voltage. In Fig. 13(a), the observation points for titanium ions are varied from the target surface (distance: 0 mm) to the near-edge of the solenoid coil (distance: 68.8 mm). The argon spectra shown in Fig. 13(b) are observed at the surface of the target (distance: 0 mm). The x-axis in both figures indicates the time after the pulsed voltage is applied to the sputter target; time "0" is when the RF voltage is applied. The initial source voltage applied to the target was -0.6 kV.

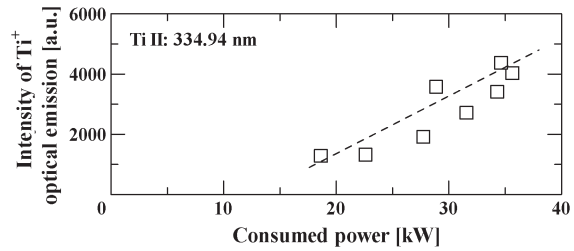

Fig. 15. Intensity of the optical emission spectrum of singly ionized titanium optical emission spectrum (wavelength: 334.94 nm) as a function of the power consumed in the target plasma (the experimental conditions are same as Fig. 10, and delay time: 100 μs).

The emission intensity of titanium ion spectra initially increases, corresponding to a buildup of the ion species. The intensity is saturated at approximately 10 μs. However, the build-up time for ion production is different for titanium and argon species because argon ions have been already produced by the RF plasma prior to the application of the pulse voltage. Therefore, the argon ion optical emission is recognized at the instant that the pulse voltage is applied to the target. Conversely, the emission intensity for the titanium ions gradually increases based on the pulse voltage application. The saturated state tends to be delayed as the distance from the sputter target increases.

As shown in Fig. 13, the titanium ions are produced in a region approximately 10 mm from the target surface and gradually diffuse into the RF plasma. As a result of this behavior, the emission intensity at 17.2 mm gradually increases with time. The emitted titanium ions are distributed within a limited range from 8.6 to 34 mm. Beyond 10 mm, the intensity at 34 mm decreases to 25% of the value observed at the target surface (see Fig. 14). When the application of the pulse voltage is terminated, the emission intensity swiftly decreases with time, reaching approximately 20% of the original intensity at 40 μs, which is only 10 μs after the end of the pulse. The production of argon ions is facilitated when the pulse voltage is applied to the target, and the intensity gradually increases due to the target plasma production. Thus, the application of the pulse voltage produces more argon ions, and the emission intensity increases with time.

The titanium ion emission intensity (334.94 nm) as a function of the power consumed in the target plasma is shown in Fig. 15. A linear relationship between the consumed power and the emission intensity was observed.

The behavior of the emission intensity was also investigated for a variety of RF voltages, and the results are shown in Fig. 16. All of the species were observed to similarly behave, i.e., the emission intensity increases as the RF voltage increases from

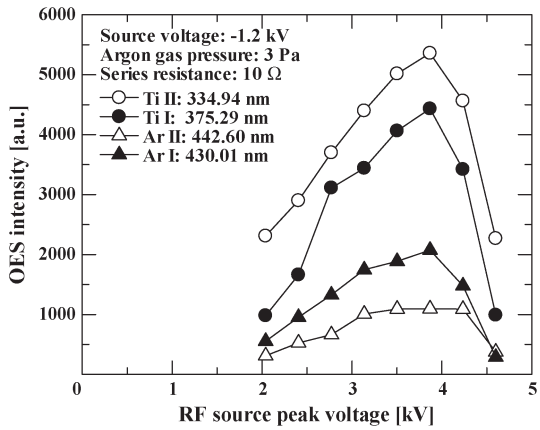

Fig. 16. Intensities of optical emission spectra of titanium and argon species at the target surface as a function of the 200-kHz RF source voltage (the experimental conditions are identical to Fig. 11).

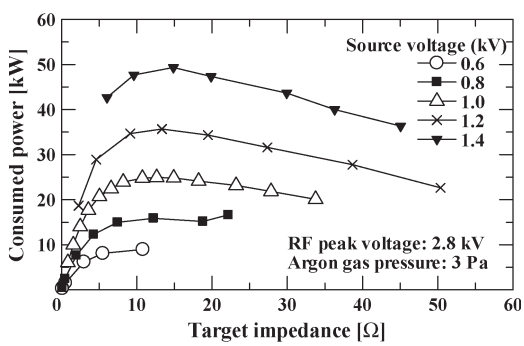

Fig. 17. Power consumed in the target plasma as a function of the target impedance for target voltages of −0.6, −0.8, −1, −1.2, and −1.4 kV (RF source voltage: 2.8 kV (peak value); and argon gas pressure: 3 Pa).

2 to 3.6 kV, which is caused by the increased power consumed in the target plasma, and the peak intensity occurred at an RF voltage of 4 kV. The intensity decreases for RF voltages over approximately 4 kV because the power decreases, as shown in Fig. 10.

To evaluate the relationship between the emission intensity and the electrical parameters of the target plasma, the behavior of the titanium ion emission intensity is discussed on the basis of impedance matching.

The consumed power and the titanium ion emission intensity as a function of the impedance of the target plasma are shown in Figs. 17 and 18, respectively. When the source voltage applied to the target is varied from 0.6 to 1.4 kV with negative polarity, the power commonly has a peak maximum at an impedance value of approximately 10 Ω, which results in impedance matching of the circuit for R1 at 10 Ω. The emission intensity shown in Fig. 18 also peaks at an impedance value of 10 Ω. Thus, the emission intensity has a peak maximum at the point of impedance matching.

The sputtering yield as a function of the target voltage is shown in Fig. 19. It is clear that the sputtering yield increases as the target voltage increases [14]. Fig. 20 shows the product of the plasma density and the sputtering yield as a function of the plasma impedance. The plasma density is derived from Fig. 9, and the sputtering yield is obtained using Figs. 10 and 19.

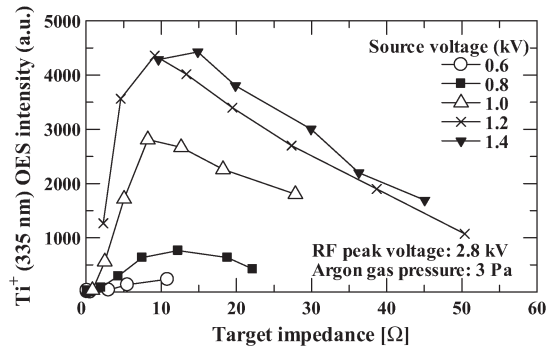

Fig. 18. Intensities of optical emission spectra of singly ionized titanium (334.94 nm) as a function of the target impedance for target voltages of −0.6, −0.8, −1, −1.2, and −1.4 kV (RF source voltage: 2.8 kV (peak value); and argon gas pressure: 3 Pa).

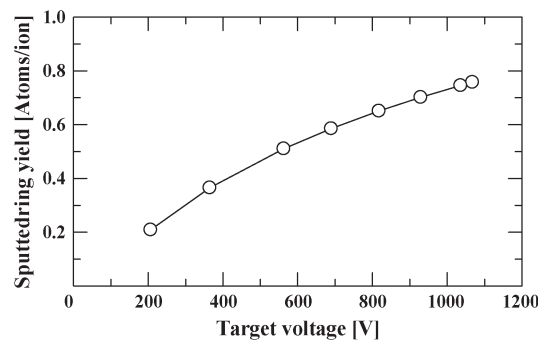

Fig. 19. Sputtering yield as a function of the target voltage.

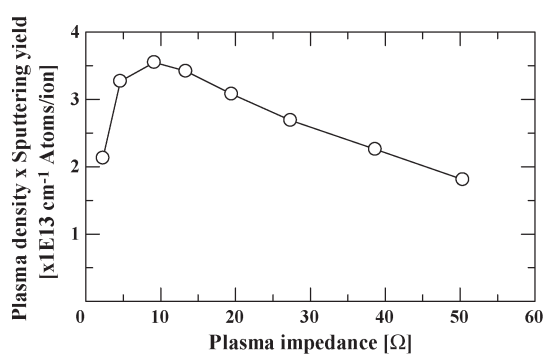

Fig. 20. Product of plasma density and sputtering yield as a function of the plasma impedance. The plasma density is derived from Fig. 9, and the sputtering yield is derived from Fig. 19.

The plasma density is closely related to the current, and the sputtering yield is closely related to the target voltage for ion acceleration. The argon plasma density ranges from 2 to 7×10^{13} cm^{-3} for RF voltages from 2 to 4 kV and an argon gas pressure of 1 Pa. These values were obtained using a Langmuir probe experiment. Thus, the target is efficiently sputtered due to a synergistic effect of the ion density and the sputtering yield, and the product of both parameters has a peak maximum at impedance matching conditions. Fig. 21 shows the emission intensity by titanium species as a function of the consumed power for a variety of target voltages with negative polarity. Although a trend of increasing the emission intensity with the consumed power is observed, the intensity slightly changes for the voltages ranging from 0.6 to 1.2 kV, and the intensity for an

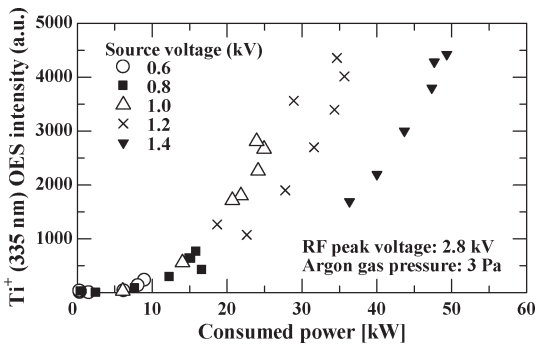

Fig. 21. Intensities of optical emission spectra of singly ionized titanium (334.94 nm) as a function of the power consumed in the target plasma for target voltages of −0.6, −0.8, −1, −1.2, and −1.4 kV (RF source voltage: 2.8 kV (peak value); and argon gas pressure: 3 Pa).

applied voltage of 1.4 kV is lower than those observed for the lower applied voltages. To clarify the relationship between the emission intensity and the power, more parametric surveys on target plasma formation should be conducted.

IV. Conclusion

A high-power pulsed glow plasma has been generated using a pulsed RF (200-kHz) ICP. The RF originated from the resonance between a 24-nF capacitor and a 26-μH inductor. The RF signal was supplied in a burst mode that lasted 200–260 μs. A pair of titanium electrodes as the sputter target (diameter, 60 mm; thickness, 10 mm) was placed near the RF plasma region, and a negatively pulsed bias at the electrodes was used to attract argon ions into the RF plasma. Bombardment of the electrodes produced abundant titanium ion species to make a plasma, which is referred to as the target plasma. Thus, the target material is sputtered and ionized in the target plasma for a duration as short as 10 μs. The particle density of the target plasma was observed to have a linear relationship to the target current.

The timing of the pulsed voltage application was adjusted based on the RF signal; thus, plasma generation was conducted by a burst pulse operation. No externally applied magnetic field or impedance matching circuit was used in this system. The RF plasma was initiated at approximately 30 μs after the burst voltage application, and the initiation time of the RF plasma was a minimum of approximately 26 μs at 2 Pa. The particle density of the target plasma was on the order of 10^{13}–10^{14} cm^{-3} at an argon gas pressure of 1 Pa; thus, the ionization rate reached a maximum of 10%. The instantaneous power density at the target was approximately 400 W/cm^2 at 32 kW of instantaneous power. This high density is due to the high current through the inductor, which reached a maximum of 100 A for an RF burst voltage of approximately 3 kV. When the RF plasma was terminated, the density of the target plasma decreased to approximately 10% of the original value due to a decrease in the argon ion concentration via deionization of the charged particles.

The optical emission spectra were observed both to confirm the ionization of the sputtered metallic species and to investigate the ion distribution in the plasma space. The optical emission spectra confirmed that the target material was sputtered and ionized to titanium ion species and distributed in the RF plasma region. The titanium ions are mainly limited to the region within approximately 50 mm of the sputter target. The ion behavior was temporally dependent in the RF plasma region, which differed from that of the argon ions because the argon ions are generated by the RF plasma prior to the target plasma initiation; however, the target plasma facilitates argon ion production.

The emission intensity of the titanium ions was observed to have a linear relationship with the power consumed in the target plasma. The emission intensity exhibited a peak maximum at a target plasma impedance of 10 Ω when the RF voltage was varied; thus, the emission intensity exhibits a peak maximum under conditions of impedance matching of the target current circuit. The intensity was found to be dependent on the plasma density and the sputtering yield.

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